

E-COMMERCE REVENUE & CUSTOMER BEHAVIOUR ANALYTICS

End-to-End E-Commerce Analytics Project

A comprehensive analytics solution covering data exploration, cleaning, feature engineering, SQL-driven analytical queries, PostgreSQL integration, and Power BI dashboard development.

Dataset: Customer Shopping Behaviour • 3,900 Transactions • 19 Features

Tools: Python • PostgreSQL • SQLAlchemy • Power BI

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1. Executive Summary

This report presents an end-to-end analytics project analysing customer purchasing behaviour and revenue patterns for an e-commerce retail company. The project encompasses the full data analytics pipeline — from raw dataset ingestion and Python-based exploratory data analysis through to SQL-driven data analytics and an interactive Power BI dashboard.

The dataset comprises 3,900 customer transactions spanning four product categories across all seasons, with rich attributes covering demographics, payment behaviour, shipping preferences, and subscription status. Two engineered features were introduced during preprocessing to enrich the analytical depth. The resulting insights address ten structured analytical questions formulated as part of this project and are summarised in an interactive Power BI Customer Behaviour Dashboard.

1.1 Key Performance Indicators

Total Customers	Average Purchase Amount	Total Revenue	Average Review Rating
3900	\$59.76	\$233,081	3.75 / 5.0

1.2 Critical Findings

- Clothing is the dominant revenue category at \$104,264 (44.7% of total revenue) across 1,737 transactions, followed by Accessories at \$74,200.
- Young Adult customers generate the highest revenue at \$62,143, though revenue is distributed relatively evenly across all four age groups — a sign of broad market reach.
- Male customers account for 67.9% of the customer base and 67.7% of total revenue (\$157,890), reflecting a significant demographic concentration.
- Only 27% of customers subscribe — yet non-subscribers and subscribers show near-identical average spend (\$59.87 vs \$59.49), suggesting subscription incentives have not shifted basket size.
- Approximately 80% of customers are classified as Loyal (3,116 customers with more than 10 previous purchases), indicating strong baseline retention.
- Fall is the highest-revenue season at \$60,018, while Summer trails at \$55,777 — a notable seasonal swing that should inform inventory and marketing planning.
- Electronic check usage (as seen in the churn project) has a parallel here: customers using premium or frictionless shipping options (2-Day, Express) spend slightly more per order.

2. Project Overview

2.1 Business Problem

An e-commerce company possesses substantial transaction data but lacks a centralised reporting solution capable of converting that data into decision-ready business intelligence. Management has identified four core knowledge gaps:

- Which products and categories drive the most revenue, and which underperform?
- Which customer segments are most valuable, and how do demographics influence purchasing behaviour?
- How does subscription status, discount usage, and shipping preference affect revenue?
- What seasonal and product-level trends should inform inventory planning and promotional strategy?

Ten specific analytical questions were formulated and explored as part of this project, forming the backbone of the SQL analysis component (see Section 5).

2.2 Project Objectives

- Clean, enrich, and standardise the raw shopping behaviour dataset for analytical use.
- Engineer features that add demographic segmentation and purchase frequency dimensions.
- Develop an interactive Power BI dashboard exposing KPIs, category performance, age-group revenue patterns, and subscription behaviour.
- Deliver actionable business recommendations grounded in the data.

2.3 Tools and Technologies

- Python (*Pandas*, *NumPy*, *Matplotlib*, *Seaborn*): data loading, exploration, cleaning, feature engineering, and visualisation.
- *PostgreSQL*: relational database for persistent storage and structured SQL querying.
- *SQLAlchemy* with *python-dotenv*: secure, credentials-protected database connectivity layer, pushing the cleaned DataFrame to the *revenue_db* database.
- Power BI: interactive dashboard development for the Customer Behaviour Dashboard.
- GitHub: version control and project documentation.

3. Dataset Description

3.1 Source

The dataset is a customer shopping behaviour dataset containing 3,900 transaction records across 17 original fields. One redundant field (*promo_code_used*) was identified as a perfect duplicate of *discount_applied* and dropped during preprocessing. Missing values in *review_rating* were imputed using the median rating within each product category rather than a global median, preserving category-level rating variance. After cleaning, two engineered features were added, yielding a final analytical schema of 19 columns.

Column names were standardised to lowercase *snake_case* during preprocessing to align with PostgreSQL naming conventions and reduce query-writing friction.

3.2 Dataset Schema

The table below describes all 19 columns in the cleaned analytical dataset, including the two engineered features.

Column Name	Data Type	Description
<i>customer_id</i>	Integer	Unique identifier for each customer
<i>age</i>	Integer	Customer age in years
<i>gender</i>	String	Customer gender (Male / Female)
<i>item_purchased</i>	String	Name of the product purchased
<i>category</i>	String	Product category (Clothing / Accessories / Footwear / Outerwear)
<i>purchase_amount</i>	Integer	Transaction value in USD
<i>location</i>	String	Customer's state/region
<i>size</i>	String	Product size selected (S / M / L / XL)
<i>color</i>	String	Product colour selected
<i>season</i>	String	Season of purchase (Spring / Summer / Fall / Winter)
<i>review_rating</i>	Float	Customer review score (2.5–5.0); NA values imputed by category median
<i>subscription_status</i>	String	Whether the customer holds a subscription (Yes / No)
<i>shipping_type</i>	String	Shipping method chosen (Free Shipping / Standard / Express / etc.)
<i>discount_applied</i>	String	Whether a discount was applied to the transaction (Yes / No)

<i>previous_purchases</i>	Integer	Count of prior purchases by the customer
<i>payment_method</i>	String	Payment instrument used (PayPal / Credit Card / Cash / Debit Card / Venmo / Bank Transfer)
<i>frequency_of_purchases</i>	String	Self-reported purchase frequency (Weekly / Monthly / Annually / etc.)
<i>age_group</i>	String	Engineered — Quartile-based age segmentation (Young Adult / Adult / Middle Age / Senior)
<i>purchase_frequency_days</i>	Integer	Engineered — Numeric translation of frequency_of_purchases into days

3.3 Dataset Statistics

- Total records: 3,900 transactions
- Gender split: Male 2,652 (68%) | Female 1,248 (32%)
- Category split: Clothing 44.5% | Accessories 31.8% | Footwear 15.4% | Outerwear 8.3%
- Subscription split: Non-subscriber 73% | Subscriber 27%
- Discount applied: Yes 43% | No 57%
- Purchase amount range: \$20 – \$100 (Mean: \$59.76, Median: \$60.00)
- Review rating range: 2.5 – 5.0 (Mean: 3.75)
- Season distribution: approximately equal across all four seasons (~25% each)

4. Methodology & Project Workflow

The project follows a structured five-phase workflow from raw data ingestion through to dashboard delivery.

Phase 1 — Data Understanding

The raw CSV (*customer_shopping_behavior.csv*) was loaded into a Pandas DataFrame and reviewed using *df.info()*, *df.describe()*, and *df.head()*. Key activities included:

- Confirming the presence of 17 original fields spanning demographics, transaction details, and behavioural attributes.
- Identifying *review_rating* as having null values requiring imputation.
- Confirming *promo_code_used* is a binary mirror of *discount_applied* using a vectorised equality check: *(df['discount_applied'] == df['promo_code_used']).all()* returned True, justifying the column's removal.
- Reviewing *purchase_amount_(usd)* for any outliers or negative values — none were found.

Phase 2 — Data Cleaning

Four specific cleaning operations were performed:

- Missing value imputation: *review_rating* null values were filled using the per-category median via *groupby().transform(lambda x: x.fillna(x.median()))*. This approach avoids the distortion of applying a single global median to products with systematically different rating profiles.
- Column name standardisation: all column names were converted to lowercase and spaces replaced with underscores (*str.lower()*, *str.replace(' ', '_')*) for SQL compatibility.
- Column rename: *purchase_amount_(usd)* was renamed to *purchase_amount* to remove special characters.
- Redundant column removal: *promo_code_used* was dropped as it was confirmed to be a duplicate of *discount_applied*.

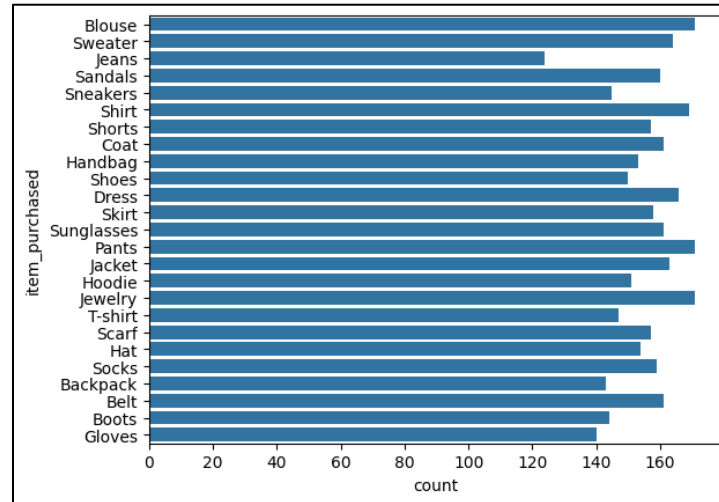
The cleaned dataset was exported to *Updated_Revenue_Dataset.csv* and pushed to PostgreSQL.

Phase 3 — Exploratory Data Analysis (EDA)

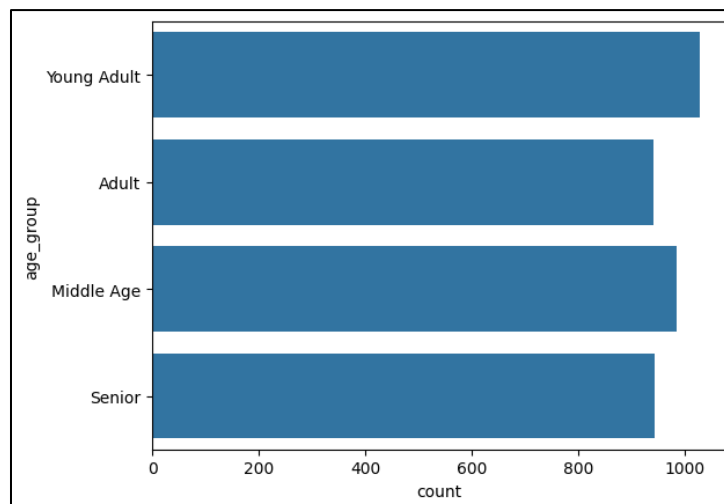
Eleven plots were produced in seaborn to build a working understanding of the data before SQL querying and dashboard development. The plots are organised below by analytical type.

Univariate Analysis

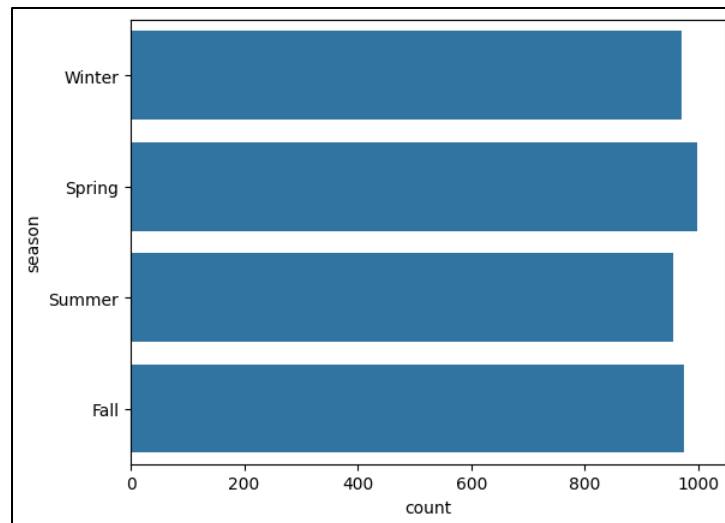
- **Count Plot** — The distribution is notably flat — Blouse, Pants, and Jewelry each record 171 transactions while the lowest-count items are not far behind. No single SKU dominates, which indicates healthy product spread but also limits the ability to concentrate marketing on a few blockbuster items.



- **Count Plot** — Validates the output of the *age_group* binning by confirming roughly equal group sizes — Young Adult (1,028), Middle Age (986), Senior (944), and Adult (942). The near-uniform distribution across age groups confirms that quartile binning produced balanced segments and that no age cohort is structurally over- or under-represented in the customer base.

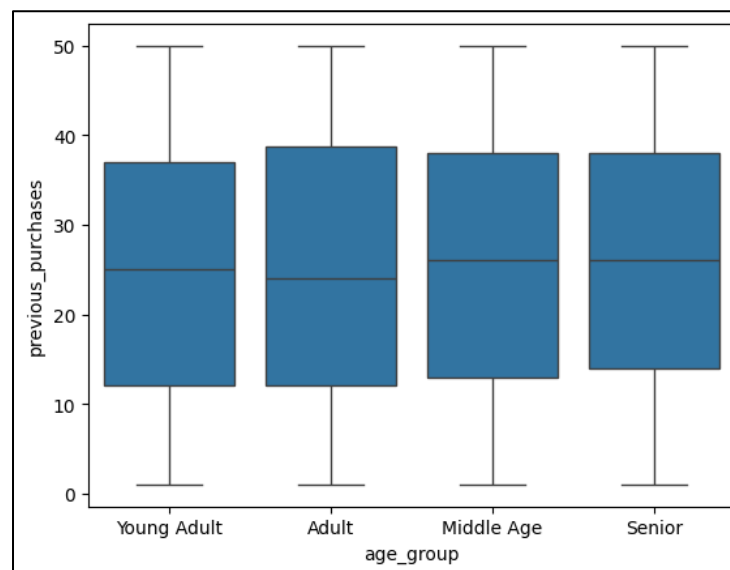


- **Count Plot** — Displays the number of transactions recorded in each season. Spring leads narrowly at 999 transactions, followed by Fall (975), Winter (971), and Summer (955). The near-equal split confirms that the dataset captures purchases throughout the year without a strong seasonal concentration in raw volume — though revenue per transaction varies, as explored later in the SQL analysis (Fall: \$60,018; Summer: \$55,777).



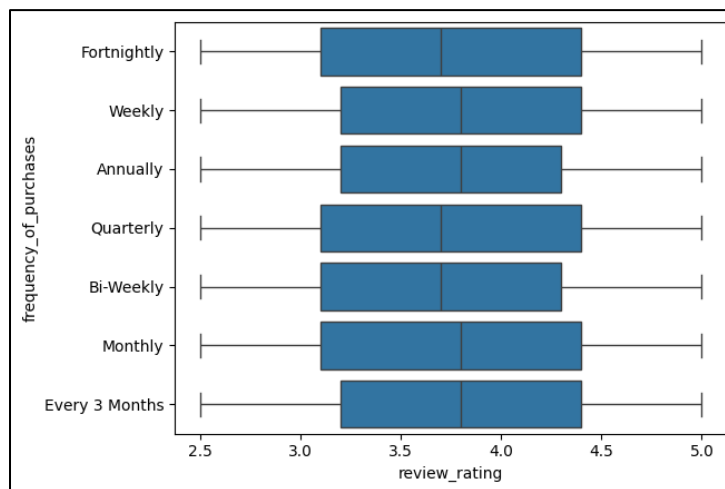
Bivariate Analysis

- **Box Plot — *age_group* vs. *previous_purchases*** : Examines whether prior purchase count varies across age segments. The median and interquartile range of *previous_purchases* is broadly consistent across all four age groups, indicating that loyalty behaviour is not age-driven in this dataset. All groups include high-frequency buyers (visible as upper outliers), confirming that the Loyal segment identified in Q7 spans the full age spectrum rather than concentrating in any one cohort.

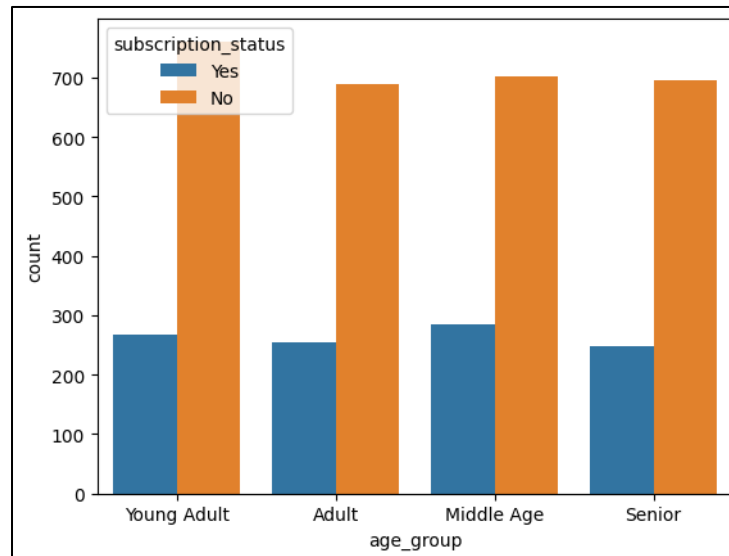


- **Box Plot — *review_rating* vs. *frequency_of_purchases*** : Tests whether how frequently a customer shops correlates with their satisfaction ratings. Across all seven purchase frequency categories (Weekly, Fortnightly, Bi-Weekly, Monthly, Every 3 Months, Quarterly, Annually), the median review rating and interquartile spread remain near-identical. This indicates that satisfaction is driven by product and service quality rather than by how often a customer transacts, and validates that the category-median

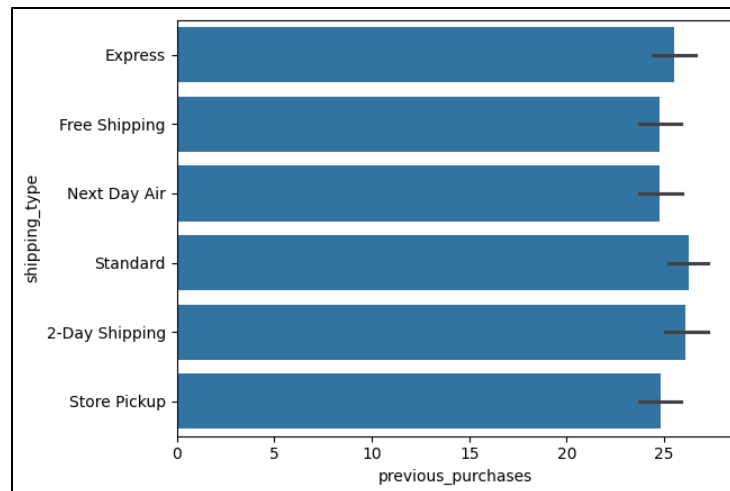
imputation applied to missing *review_rating* values does not introduce frequency-related bias.



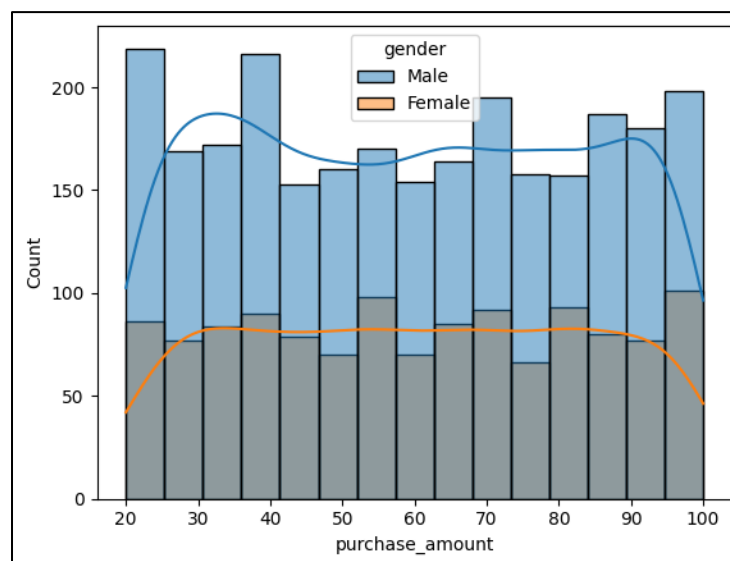
- Count Plot — *age_group* by *subscription_status*** : A grouped bar chart showing how subscription adoption is distributed across the four age groups. Non-subscribers dominate in every group (roughly 73% of each), and subscription counts are also broadly consistent across Young Adult, Adult, Middle Age, and Senior segments. This reveals that low subscription penetration (27% overall) is not concentrated in any particular age cohort — it is a systemic pattern, not a demographic one. This finding directly motivates the subscription monetisation recommendations in Section 7.



- Bar Plot — *previous_purchases* vs. *shipping_type*** : Plots mean previous purchases for each shipping method to explore whether veteran buyers prefer specific delivery options. Average prior purchase counts are near-uniform across all six shipping types (Free Shipping, Standard, Store Pickup, Next Day Air, Express, 2-Day Shipping), indicating that shipping preference is not determined by customer seniority. Customers select shipping based on immediacy or cost rather than their history with the company.



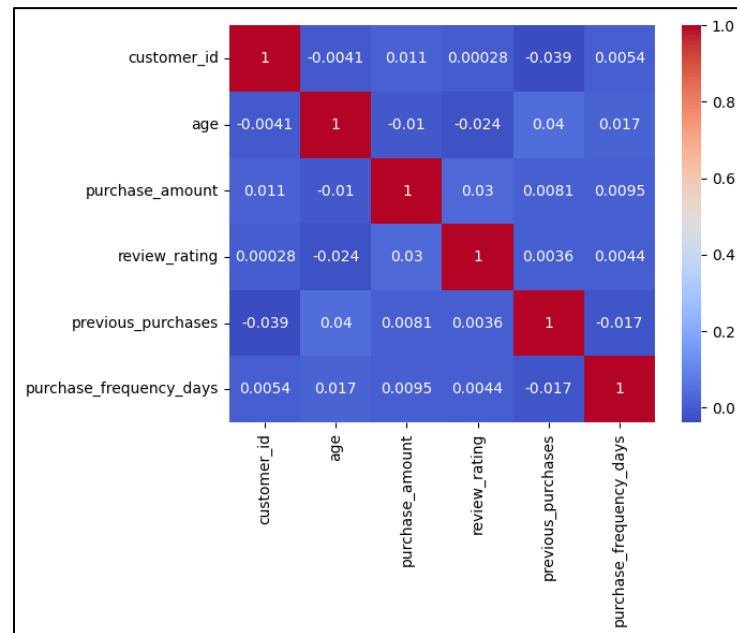
- Histogram — *purchase_amount* vs. *gender*** : Plots mean previous purchases for each shipping method to explore whether veteran buyers prefer specific delivery options. Average prior purchase counts are near-uniform across all six shipping types (Free Shipping, Standard, Store Pickup, Next Day Air, Express, 2-Day Shipping), indicating that shipping preference is not determined by customer seniority. Customers select shipping based on immediacy or cost rather than their history with the company.



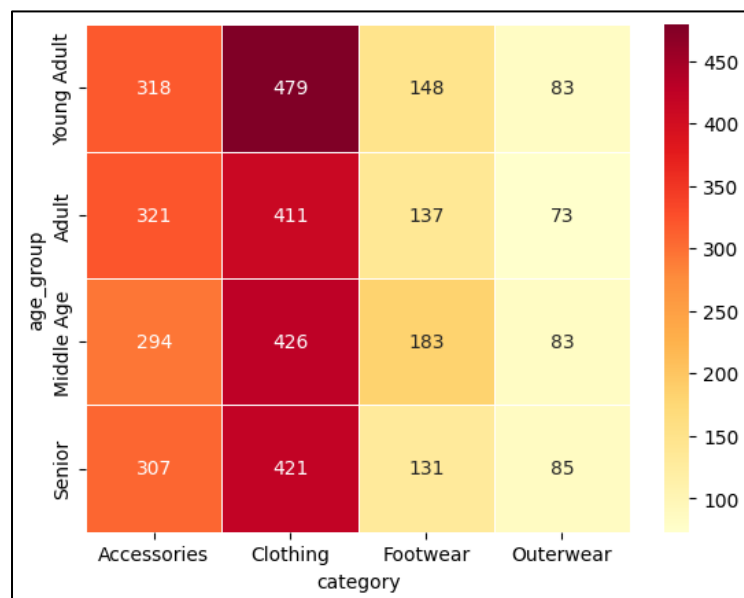
Multivariate Analysis

- Correlation Heatmap — numeric features** : A full annotated correlation matrix across all numeric columns — *age*, *purchase_amount*, *review_rating*, *previous_purchases*, and *purchase_frequency_days*. The heatmap confirmed weak correlations across all pairs (coefficients close to 0), with no strong linear relationship between any two numeric variables. Notably, *purchase_amount* shows near-zero correlation with *age*, *previous_purchases*, and *review_rating* — meaning spend is not determined by age, loyalty level, or satisfaction in any simple linear way. This directed attention toward

categorical variables (contract type, subscription status, payment method) as the primary segmentation levers, which the SQL analysis in Section 5 subsequently confirms.



- Correlation Heatmap — *age_group* vs. *category*** : Displays the transaction count across combinations of customer age group and product category, highlighting purchasing patterns between demographic segments and product types. Clothing records the highest transaction counts across every age group, followed by Accessories, while Footwear and Outerwear consistently account for fewer purchases. The pattern remains largely uniform across all four age groups, indicating that product preferences are broadly similar regardless of age. This reinforces earlier findings that the customer base is well distributed across demographics and that category popularity is driven by overall demand rather than age-specific purchasing behaviour.



Cross-Variable Observations

- Discount application (43% of transactions) showed negligible impact on average spend (\$59.28 discounted vs. \$60.13 non-discounted), indicating that discounts are not materially shifting basket size. Combined with the weak correlations identified in the heatmap, this finding suggests that spend behaviour in this dataset is largely independent of the levers most commonly used to influence it — discount, age, and purchase history. The primary differentiators identified through the plots are categorical: which age group a customer belongs to, whether they subscribe, and which shipping method they choose.

Phase 4 — Feature Engineering

Two features were engineered to enhance analytical utility:

- age_group*: customers were segmented into four quartile-based age groups using *pd.qcut()* with labels ['Young Adult', 'Adult', 'Middle Age', 'Senior']. Quartile-based binning was preferred over fixed-interval binning to ensure balanced group sizes across the age distribution.
- purchase_frequency_days*: the categorical *frequency_of_purchases* field was mapped to a numeric equivalent in days using a defined dictionary (e.g., Weekly → 7, Monthly → 30, Annually → 365). This conversion enables frequency-based quantitative analysis and customer lifetime value estimation.

Phase 5 — Database Integration

The cleaned and feature-engineered DataFrame was pushed to a locally hosted PostgreSQL database (*revenue_db*) using *SQLAlchemy*. Credentials were securely managed via environment variables loaded with *python-dotenv*, keeping sensitive connection details out of the notebook source code. The table *revenue_data* was created with 3,900 rows and 19 columns.

5. SQL-Driven Business Analysis

Ten analytical questions were identified and explored through structured SQL queries against the *revenue_data* table in PostgreSQL. These questions were designed to surface key business patterns across customer behaviour, product performance, and revenue distribution. A summary of all questions and findings is provided first, followed by individual query definitions.

5.1 Analysis Summary

Q#	Analysis Question	Key Finding
Q1	Revenue by Gender	Male customers generate \$157,890 vs. \$75,191 for female customers, reflecting the 68:32 customer count split.
Q2	High-Spend Discount Users	Customers who applied a discount but still spent above the dataset average — useful for identifying deal-responsive high-value buyers.
Q3	Top 5 Products by Avg Rating	Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78) — all footwear and accessories dominate quality perception.
Q4	Standard vs Express Shipping Spend	Express shipping customers spend \$60.48 on average vs. \$58.46 for standard — a modest but consistent premium.
Q5	Subscription vs Non-Subscription Spend	Non-subscribers (2,847) account for \$170,436 (73.1%) of total revenue; subscribers show similar avg spend (\$59.49 vs \$59.87).
Q6	Products with Highest Discount Rates	Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%) — nearly every other purchase uses a discount.
Q7	Customer Segmentation	Loyal: 3,116 Returning: 701 New: 83 — the customer base is overwhelmingly composed of loyal repeat buyers.
Q8	Top 3 Products per Category	Blouse/Pants/Shirt (Clothing), Jewelry/Belt/Sunglasses (Accessories), Sandals/Shoes/Sneakers (Footwear), Jacket/Coat (Outerwear).
Q9	Repeat Buyers & Subscription	Of 3,476 repeat buyers (>5 purchases), 958 subscribe and 2,518 do not — revealing a large monetisation gap.
Q10	Revenue by Age Group	Young Adult leads at \$62,143, followed by Middle Age (\$59,197), Adult (\$55,978), and Senior (\$55,763).

5.2 Query Definitions

Q1 — Revenue by Gender

Computes total purchase revenue grouped by customer gender to identify the primary revenue-generating demographic.

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM revenue_data
GROUP BY gender;
```

Gender	Customers	Total Revenue (USD)
Male	2,652	\$157,890
Female	1,248	\$75,191

Q2 — High-Spend Discount Users

Identifies customers who applied a discount but still spent above the dataset's average purchase amount — a high-value, deal-responsive segment.

```
SELECT age_group, customer_id, purchase_amount
FROM revenue_data
WHERE discount_applied = 'Yes'
      AND purchase_amount >= (SELECT AVG(purchase_amount) FROM revenue_data);
```

Q3 — Top 5 Products by Average Review Rating

Ranks products by average customer review score to identify quality leaders across the catalogue.

```
SELECT item_purchased,
       ROUND(AVG(review_rating::numeric), 2) AS "Average Product Rating"
FROM revenue_data
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;
```

Product	Avg Review Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Q4 — Standard vs Express Shipping Spend Comparison

Compares the average purchase amount between customers selecting Standard and Express shipping to test whether urgency of delivery correlates with order value.

```
SELECT shipping_type,
       ROUND(AVG(purchase_amount::numeric), 2) AS avg_spend
FROM revenue_data
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```


Express shipping customers average \$60.48 per order versus \$58.46 for Standard — a consistent \$2.02 premium. This may reflect that Express users are making more considered, higher-value purchases.

Q5 — Subscription vs Non-Subscription Revenue Comparison

Compares customer count, average spend, and total revenue between subscribed and non-subscribed customers.

```
SELECT subscription_status,  
       COUNT(customer_id) AS total_customer,  
       ROUND(AVG(purchase_amount::numeric), 2) AS avg_spent,  
       ROUND(SUM(purchase_amount::numeric), 2) AS total_revenue  
FROM revenue_data  
GROUP BY subscription_status  
ORDER BY total_revenue, avg_spent DESC;
```

Subscription Status	Customer Count	Avg Spend (USD)	Total Revenue (USD)
No	2,847	\$59.87	\$170,436
Yes	1,053	\$59.49	\$62,645

Q6 — Products with Highest Discount Application Rates

Identifies the five products where the highest proportion of transactions included a discount — useful for assessing discount dependency and margin pressure by SKU.

```
SELECT item_purchased,  
       ROUND(100 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END) /  
COUNT(*), 2) AS discount_rate  
FROM revenue_data  
GROUP BY item_purchased  
ORDER BY discount_rate DESC  
LIMIT 5;
```

Hat (50.00%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%). Half of all Hat and Sneaker purchases use a discount — pricing strategy for these items warrants review.

Q7 — Customer Segmentation by Purchase History

Segments customers into New, Returning, and Loyal tiers based on previous purchase count using a CTE.

```
WITH customer_type AS (  
  SELECT customer_id, previous_purchases,  
         CASE  
           WHEN previous_purchases = 1 THEN 'New'  
           WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'  
           ELSE 'Loyal'  
         
```

```

        END AS customer_segment
    FROM revenue_data
)
SELECT customer_segment, COUNT(*) AS number_of_customers
FROM customer_type
GROUP BY customer_segment;

```

Customer Segment	Previous Purchases Criterion	Customer Count
Loyal	> 10 purchases	3,116
Returning	2 – 10 purchases	701
New	1 purchase	83

Q8 — Top 3 Products per Category (Window Function)

Uses ROW_NUMBER() as a window function partitioned by category to return the three most-purchased products within each product category.

```

WITH item_counts AS (
    SELECT category, item_purchased,
           COUNT(customer_id) AS total_orders,
           ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id)
                               DESC) AS item_rank
    FROM revenue_data
    GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;

```

Item Rank	Category	Items Purchased	Total Orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Q9 — Repeat Buyers and Subscription Affinity

Examines whether customers with more than five previous purchases are likely to also hold an active subscription — testing whether purchase loyalty and subscription adoption are correlated.

```
SELECT subscription_status, COUNT(customer_id) AS repeat_buyers
FROM revenue_data
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Of 3,476 repeat buyers with more than 5 previous purchases, 2,518 (72.4%) are non-subscribers and only 958 (27.6%) subscribe. This mirrors the overall subscription rate, indicating that repeat purchase frequency does not strongly predict subscription adoption. This is a significant monetisation opportunity.

Q10 — Revenue by Age Group

Aggregates total revenue by the engineered age_group feature to identify the highest-spending demographic cohort.

```
SELECT age_group, SUM(purchase_amount) AS total_revenue
FROM revenue_data
GROUP BY age_group
ORDER BY total_revenue DESC;
```

Age Group	Customers	Total Revenue (USD)
Young Adult	1,028	\$62,143
Middle Age	986	\$59,197
Adult	942	\$55,978
Senior	944	\$55,763

6. Power BI Dashboard

The Power BI Customer Behaviour Dashboard was built using the cleaned CSV (*Updated_Revenue_Dataset.csv*) as its data source. The dashboard is designed for business stakeholders and presents customer, revenue, and product insights in a single-page interactive layout with sidebar filter controls.

6.1 Dashboard Layout Overview



6.2 KPI Cards

Three headline KPIs are presented at the top of the dashboard:

- Total Customers — 3.9K: total transaction records in the dataset, representing the customer population.
- Average Purchase Amount — \$59.76: the mean transaction value across all 3,900 records.
- Average Review Rating — 3.75: the mean customer satisfaction score, on a scale of 2.5 to 5.0.

6.3 Subscription Status Distribution

A donut chart presents the subscription split: 73% non-subscribers (gold/yellow) and 27% subscribers (dark navy). This visualisation immediately surfaces the low subscription penetration

rate and invites analytics questions about the subscription value proposition — particularly given that subscribed and non-subscribed customers spend at nearly identical rates.

6.4 Sales and Revenue by Category

Two adjacent bar charts compare category performance on volume and value dimensions:

Sales by Category

- Clothing: ~1,700 orders (highest)
- Accessories: ~1,250 orders
- Footwear: ~600 orders
- Outerwear: ~325 orders

Revenue by Category

- Clothing: ~\$104K (highest)
- Accessories: ~\$74K
- Footwear: ~\$36K
- Outerwear: ~\$18K

The volume and revenue rankings are consistent across both charts, indicating no significant unit-price anomalies between categories — all categories have comparable average selling prices around \$55–60.

6.5 Revenue and Sales by Age Group

Two horizontal bar charts in the bottom row break down revenue and order count by age group:

Revenue by Age Group

Age Group	Customers	Total Revenue (USD)
Young Adult	1,028	\$62,143
Middle Age	986	\$59,197
Adult	942	\$55,978
Senior	944	\$55,763

Young Adults contribute the highest revenue (\$62,143), but the distribution across all four age groups is notably even — ranging from \$55,763 (Senior) to \$62,143 (Young Adult), a spread of only \$6,380. This suggests the product range has genuine cross-generational appeal and that no single age cohort is being over- or under-served.

Sales by Age Group

The Sales by Age Group chart mirrors the revenue pattern — Young Adults lead on order count as well, followed closely by Middle Age, Senior, and Adult. The near-equal distribution across age segments suggests even customer acquisition across demographics.

7. Key Findings & Business Recommendations

7.1 Finding 1 — Clothing Dominates but Outerwear Needs Attention

Clothing accounts for 44.7% of total revenue (\$104,264) with 1,737 transactions. Outerwear, by contrast, generates only \$18,524 from 324 transactions — the smallest category by both volume and revenue.

Recommendations

- Investigate whether Outerwear's low volume reflects limited product range, higher price points, or seasonal suppression. If it is seasonal, align inventory and promotional intensity with Fall and Winter peaks.
- Cross-sell Outerwear with Clothing purchases during Fall — the highest-revenue season — through bundling or 'complete the look' prompting.
- Evaluate whether expanding the Footwear range (currently third by revenue) could grow its contribution, given it has the highest average review rating (3.79) of any category.

7.2 Finding 2 — Subscription Penetration is Low Despite Loyal Customer Base

Only 27% of customers subscribe, yet the customer base is overwhelmingly loyal — 80% have made more than 10 previous purchases. Average spend between subscribers and non-subscribers is nearly identical (\$59.49 vs \$59.87). This means the subscription programme is not driving incremental spend.

Recommendations

- Reposition the subscription programme around perks beyond discounts — for example, early product access, free expedited shipping, or exclusive product lines — to make subscription feel like a status upgrade rather than a discount mechanism.
- Target the 2,518 repeat buyers (>5 purchases) who are not subscribed with a personalised upgrade offer. This group has already demonstrated loyalty and is the highest-potential conversion cohort.
- Conduct an A/B test offering a complimentary 3-month trial subscription to non-subscribed loyal customers to measure whether trial access converts to paid subscription.

7.3 Finding 3 — Discount Strategy May Be Undermining Margin on Key Products

Half of all Hat and Sneaker transactions include a discount (50% and 49.66% respectively). Nearly half of Coat (49.07%) and Sweater (48.17%) purchases are also discounted. At this penetration rate, customers may have been conditioned to wait for promotions before purchasing these items.

Recommendations

- Conduct a margin impact analysis on the five highest-discount products. If contribution margins are acceptable, the current strategy may be functioning as intended. If not, introduce discount caps or time-limited promotional windows rather than always-on availability.
- Test selective discount withdrawal on Hat and Sneakers for one month and monitor volume elasticity. If sales hold, the discount was unnecessary.
- Consider shifting discounts toward Free Shipping incentives (which cost less margin per transaction than product-price discounts) as an alternative conversion lever.

7.4 Finding 4 — Fall is the Revenue Peak; Summer is the Trough

Fall generates the highest revenue at \$60,018, while Summer trails at \$55,777 — a difference of \$4,241 or approximately 7.6%.

Recommendations

- Amplify Fall campaign spend, as this is the natural purchase peak. Ensure Clothing and Outerwear inventory is at maximum availability entering September.
- Investigate Summer underperformance. Is it a category-mix issue (fewer Outerwear purchases in Summer) or a demand problem? If Footwear and light Clothing are summer-relevant, consider summer-focused product promotions.
- Use seasonal revenue data to build a forward-looking quarterly revenue forecast model.

7.5 Finding 5 — Revenue is Broadly Even Across Age Groups

The four age groups contribute between \$55,763 and \$62,143 in revenue — a range of only 11.4%. This near-equal distribution suggests the product range resonates across all life stages.

Recommendations

- Rather than over-indexing marketing spend toward Young Adults (the slight leader), maintain broad demographic reach through diverse content and channel mix.
- Analyse which specific products over-index with each age group to enable more precise personalisation and recommendation engine inputs.
- Explore whether the Senior age group (currently the lowest at \$55,763) has lower average spend per transaction or lower frequency — and whether targeted product curation could close this gap.

7.6 Finding 6 — Male Customers Dominate but the Female Segment is Underserved

Male customers represent 68% of the customer base and 67.7% of revenue. However, female customers spend at a comparable average rate (\$60.24 vs \$59.56) when normalised by count. The purchase amount histogram (Phase 3 EDA) provides further confirmation of this — the KDE distributions for male and female customers are near-identical across the full \$20–\$100 range, visually ruling out spend-level differences as a contributing factor to the revenue gap.

Recommendations

- Investigate whether the gender imbalance reflects the product mix or a marketing channel bias. If the product range includes items with gender-neutral or female-skew appeal, acquisition campaigns may not be reaching this audience effectively.
- Evaluate whether specific categories (e.g., Accessories) show stronger female purchase rates, and consider targeted campaigns to grow the female customer base in those categories.

8. Project Architecture

The project follows a clean, layered pipeline architecture:

- Raw Data Source: Customer Shopping Behaviour CSV (*customer_shopping_behavior.csv*).
- Python EDA Layer (Jupyter Notebook — *Revenue_EDA.ipynb*): data loading, null imputation (category-median for *review_rating*), column standardisation, redundant column removal, feature engineering (*age_group*, *purchase_frequency_days*), EDA visualisations, and export of the cleaned CSV (*Updated_Revenue_Dataset.csv*).
- SQLAlchemy Bridge: the cleaned DataFrame is pushed to PostgreSQL using *engine.to_sql()* with *if_exists='replace'*. Database credentials are secured via environment variables using *python-dotenv*.
- PostgreSQL Database (*revenue_db*): hosts the primary analytical table *revenue_data* (3,900 rows, 19 columns). Ten structured SQL analytical queries are executed directly against this table.
- Power BI Dashboard: the Customer Behaviour Dashboard connects to the cleaned CSV and presents KPI cards, category and age-group charts, and subscription distribution visualisations with interactive filter controls.

This architecture separates data preparation (Python), structured querying (SQL), and visual reporting (Power BI) into independent, maintainable layers. Adding new data, queries, or dashboard pages requires changes to only one layer without disrupting the others.

9. Limitations & Future Scope

9.1 Limitations

- No time dimension: the dataset lacks transaction dates, preventing time-series trend analysis, month-over-month revenue tracking, or cohort retention measurement.
- No geographic granularity in Power BI: although a location column exists, geographic visualisation was not incorporated into the current dashboard.
- No profitability data: the dataset records purchase amount but not cost of goods sold, preventing gross margin or profitability analysis.
- No causal inference: the analysis is descriptive and correlational. Relationships identified (e.g., Express shipping and higher spend) cannot be claimed as causal without controlled experimentation.

9.2 Future Scope

- Predictive Subscription Model: build a binary classifier predicting which non-subscribed customers are most likely to convert, using *previous_purchases*, *purchase_frequency_days*, and category preference as features.
- Market Basket Analysis: apply association rule mining (Apriori or FP-Growth) to identify co-purchase patterns and power cross-sell recommendation engines.
- Customer Lifetime Value (CLV) Estimation: integrate the *purchase_frequency_days* feature with average order value to produce a forward-looking CLV estimate per customer.
- Geographic Revenue Map: build a Power BI map visual using the location field to identify high and low-performing states or regions.
- Seasonal Demand Forecasting: if transaction dates are available in future data, build a forecasting model for seasonal revenue planning.

10. Conclusion

This project demonstrates a complete analytics pipeline applied to a real-world e-commerce business problem. Starting from a raw customer transaction dataset, the workflow progresses through rigorous data cleaning, thoughtful feature engineering, hypothesis-driven SQL analysis, and an interactive Power BI dashboard to produce a coherent set of evidence-backed business insights.

The central findings reveal a business with a strong and loyal customer base — over 80% of customers are classified as Loyal by purchase history — yet one with clear structural opportunities. The subscription programme reaches only 27% of customers and does not appear to drive incremental spend. Key apparel products are being heavily discounted at rates approaching 50%, raising margin concerns. The Outerwear category significantly trails the rest of the range in both volume and revenue. And despite broad demographic reach across age groups and genders, female customers remain underrepresented in the customer base.

The recommendations provided in Section 7 address each of these gaps with specific, testable interventions. The project architecture — Python for data preparation, PostgreSQL for structured analysis, and Power BI for presentation — is scalable and reusable, enabling this solution to serve as a living reporting framework as new transaction data becomes available.

Appendix

A. Python Libraries Used

- pandas: data loading, wrangling, column standardisation, and feature engineering.
- numpy: numerical operations and array handling.
- matplotlib & seaborn: EDA visualisations (count plots, box plots).
- sqlalchemy: PostgreSQL connection management and DataFrame upload via `create_engine()` and `to_sql()`.
- python-dotenv: secure environment variable management for database credentials.

B. PostgreSQL Configuration

- Database: `revenue_db`
- Host: `127.0.0.1` (localhost)
- Port: `5432` (default PostgreSQL port)
- Primary table: `revenue_data` (3,900 rows, 19 columns)
- Connection driver: `psycopg2` via `SQLAlchemy`
- Credential management: password loaded from `.env` file via `python-dotenv`

C. Power BI Dashboard Components

- KPI Cards (3): Total Customers, Average Purchase Amount, Average Review Rating.
- Donut Chart: Subscription Status distribution (Yes / No).
- Bar Charts (2): Sales by Category and Revenue by Category (vertical, by category).
- Horizontal Bar Charts (2): Revenue by Age Group and Sales by Age Group.
- Sidebar Filters (4): Subscription Status, Gender, Category, Shipping Type.

D. Seasonal Revenue Summary

Season	Total Revenue (USD)
Fall	\$60,018
Spring	\$58,679
Winter	\$58,607
Summer	\$55,777

E. Shipping Type vs Average Purchase Amount

Shipping Type	Avg Purchase Amount (USD)
2-Day Shipping	\$60.73
Express	\$60.48
Free Shipping	\$60.41
Store Pickup	\$59.89
Next Day Air	\$58.63
Standard	\$58.46

F. Dataset Source

Customer Shopping Behaviour Dataset — available at Github:
https://github.com/amlanmohanty1/customer-trends-data-analysis-SQL-Python-PowerBI/blob/main/customer_shopping_behavior.csv